

THE RANKER'S MASTERBOOK

TRIG.

The Circular Arsenal

The Problem Set

106

PROBLEMS

10

CHAPTERS

3–5

DIFFICULTY

sin

106 original trigonometry problems where the identity is only half the battle – for the very top of JEE Advanced and the Mathematical Olympiad. Solutions in the companion volume.

Contents

Ten chapters, 106 problems.

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CHAPTER I

Conditional Angle Identities

Constrained angles turn symmetric sums into identities.

№1 The Tangent Product Lock

●●●●● L3

EVALUATE

If $A + B + C = \pi$, evaluate $\frac{\tan A + \tan B + \tan C}{\tan A \tan B \tan C}$.

angle-sum tangent symmetric

№2 Half-Angle Gate

●●●●● L3

EVALUATE

If $A + B + C = \pi$ and $x = \tan \frac{A}{2}$, $y = \tan \frac{B}{2}$, $z = \tan \frac{C}{2}$, find $xy + yz + zx$.

half-angle constraint tangent

№3 The Cotangent Mirror

●●●●● L3

EVALUATE

If $A + B + C = \pi$, evaluate $\cot A \cot B + \cot B \cot C + \cot C \cot A$.

cotangent triangle symmetric

№4 Cosine Squares With a Hidden Constant

●●●●● L4

EVALUATE

If $A + B + C = \pi$, simplify $\cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C$.

cosine identity triangle

№5 Sine Squares Minus the Cubic

●●●●● L4

EVALUATE

If $A + B + C = \pi$, simplify $\sin^2 A + \sin^2 B + \sin^2 C - 2 \cos A \cos B \cos C$.

sine-square cosine triangle

№6 Double-Angle Collapse

●●●●● L4

SIMPLIFY

If $A + B + C = \pi$, simplify $\sin 2A + \sin 2B + \sin 2C$.

double-angle product-to-sum triangle

№7 Equilateral Product Ceiling

●●●●● L4

FIND THE MAXIMUM

For $A, B, C > 0$ with $A + B + C = \pi$, find the maximum of $\sin A \sin B \sin C$.

maximum

Jensen

triangle

№8 Half-Tangent Sum Floor

●●●●● L4

FIND THE MINIMUM

If $A, B, C > 0$, $A + B + C = \pi$, find the minimum of $\tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2}$.

half-angle

minimum

uvw

№9 Cosine Sum Ceiling

●●●●● L4

FIND THE MAXIMUM

For $A, B, C > 0$ with $A + B + C = \pi$, find the maximum of $\cos A + \cos B + \cos C$.

maximum

cosine

triangle

№10 A Determinant That Refuses to Vary

●●●●● L5

EVALUATE

If $A + B + C = \pi$, evaluate $\begin{vmatrix} 1 & \cos A & \sin A \\ 1 & \cos B & \sin B \\ 1 & \cos C & \sin C \end{vmatrix}^2$.

determinant

half-angle

vandermonde

CHAPTER II

Product-to-Sum Telescopes



Product-to-sum and adjacent differences leave only endpoints.

№11 Cotangent Difference Chain

●●●●● L3

EVALUATE

$$\sum_{k=1}^n \frac{1}{\sin kx \sin(k+1)x}$$

telescoping cotangent finite-sum

№12 Dyadic Cosine Product

●●●●● L3

EVALUATE

$$\prod_{k=0}^{n-1} \cos(2^k x)$$

dyadic product double-angle

№13 Sine Arithmetic Progression

●●●●● L3

EVALUATE

$$\sum_{k=1}^n \sin kx$$

finite-sum product-to-sum progression

№14 Cosine Arithmetic Progression

●●●●● L3

EVALUATE

$$\sum_{k=1}^n \cos kx$$

finite-sum cosine progression

№15 Arctangent Ladder

●●●●● L4

EVALUATE

$$\sum_{k=1}^n \tan^{-1} \frac{1}{k^2 + k + 1}$$

arctangent telescope branch

№16 Sine Ratio Collapse

●●●●● L3

SIMPLIFY

$$\prod_{k=1}^n \frac{\sin(2^k x)}{\sin(2^{k-1} x)}$$

product dyadic cancellation

No17 Secant Neighbour Sum

●●●●● L4

EVALUATE

$$\sum_{k=1}^n \frac{1}{\cos kx \cos(k+1)x}$$

tangent

secant

finite-sum

№18 Odd Sine Block

● ● ● ● ● L3

EVALUATE

$$\sum_{k=1}^n \sin((2k-1)x)$$

odd-angles

finite-sum

product-to-sum

№19 Boundary Cosines Survive

● ● ● ● ● L4

PROVE THAT

$$2 \sin \frac{x}{2} \sum_{k=1}^n \sin kx = \cos \frac{x}{2} - \cos \frac{(2n+1)x}{2}$$

proof

finite-sum

boundary

№20 Zeros of a Dyadic Product

● ● ● ● ● L4

SOLVE

For $m \geq 1$, solve $\prod_{k=0}^{m-1} 2 \cos(2^k x) = 0$ on $[0, \pi)$.

zero-counting

dyadic

equation

CHAPTER III

Unit Roots & Finite Trig Sums



Finite sums and products answered by the unit circle.

№21 The Vanishing Polygon

●●●●● L3

EVALUATE

$$\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} \quad (n > 1)$$

roots-of-unity

cosine-sum

polygon

№22 Sine Product of the Regular Polygon

●●●●● L4

EVALUATE

$$\prod_{k=1}^{n-1} \sin \frac{\pi k}{n}$$

sine-product

roots-of-unity

polynomial

№23 Second Moment of Cosines

●●●●● L3

EVALUATE

$$\sum_{k=0}^{n-1} \cos^2 \frac{2\pi k}{n} \quad (n > 2)$$

power-sum

cosine

orthogonality

№24 Fourth Moment Filter

●●●●● L4

EVALUATE

$$\sum_{k=0}^{n-1} \cos^4 \frac{2\pi k}{n} \quad (n \geq 5)$$

power-sum

fourth-power

filter

№25 Reciprocal Sine Squares

●●●●● L5

EVALUATE

$$\sum_{k=1}^{n-1} \csc^2 \frac{\pi k}{n}$$

cosecant

derivative

roots

№26 Weighted Sine Sum

●●●●● L4

EVALUATE

$$\sum_{k=1}^{n-1} k \sin \frac{2\pi k}{n}$$

weighted-sum

roots-of-unity

imaginary-part

№27 Cosine Gap Product

●●●●● L4

EVALUATE

$$\prod_{k=1}^{n-1} \left(1 - \cos \frac{2\pi k}{n} \right)$$

product

cosine

roots

№28 Odd Half-Step Energy

●●●●● L3

EVALUATE

$$\sum_{k=0}^{n-1} \sin^2 \frac{(2k+1)\pi}{2n}$$

half-step

sine-square

symmetry

№29 Squared Chord Product

●●●●● L4

EVALUATE

$$\prod_{k=1}^{n-1} \left(2 - 2 \cos \frac{2\pi k}{n} \right)$$

chords

product

unit-circle

№30 Frequency Filter

●●●●● L4

EVALUATE

$$\sum_{k=0}^{n-1} \cos \frac{2\pi mk}{n}$$

filter

orthogonality

frequency

Chebyshev & Multiple Angles



Multiple-angle polynomials turn trigonometry into algebra.

№31 The Fifth-Angle Polynomial

●●●●● L3

SIMPLIFY

$\cos 5x$ in powers of $c = \cos x$

multiple-angle polynomial Chebyshev

№32 Three Roots of a Cubic Cosine

●●●●● L4

SOLVE

$8x^3 - 6x = 1$

cubic triple-angle roots

№33 The Heptagon Cubic

●●●●● L5

PROVE THAT

$\alpha = 2 \cos \frac{\pi}{7}$ satisfies $\alpha^3 - \alpha^2 - 2\alpha + 1 = 0$.

minimal-polynomial heptagon Chebyshev

№34 Nested Radical Angle

●●●●● L4

EVALUATE

$\sqrt{2 + \sqrt{2 + \sqrt{2}}}$

nested-radical half-angle cosine

№35 Sine Five Equals Sine One

●●●●● L4

SOLVE

$\sin 5x = \sin x$ for $0 \leq x < 2\pi$

equation multiple-angle root-counting

№36 Sine Quotient Polynomial

●●●●● L4

SIMPLIFY

$\frac{\sin nx}{\sin x}$

Chebyshev-U sine-quotient recurrence

№37 Zeros of T_n

●●●●● L4

EVALUATE

$$\prod_{k=1}^n \left(x - \cos \frac{(2k-1)\pi}{2n} \right)$$

zeros

Chebyshev

product

№38 Binomial Cosine Filter

●●●●● L4

EVALUATE

$$\sum_{k=0}^n \binom{n}{k} \cos((n-2k)x)$$

binomial

complex-numbers

cosine

№39 Golden Cosine Consequence

●●●●● L4

EVALUATE

If $2 \cos x$ satisfies $t^2 - t - 1 = 0$, find $\cos 5x$.

golden-ratio

pentagon

multiple-angle

№40 Polynomial Inequality Translation

●●●●● L5

PROVE THAT

$$\cos 4x \geq 8 \cos^4 x - 8 \cos^2 x \quad \text{for all real } x.$$

inequality

multiple-angle

polynomial

CHAPTER V

Trigonometric Substitution $x = \tan \theta$ Engines

The right parameter makes radicals and ranges linear.

№41 Cosine From a Rational Parameter

●●●●● L3

FIND THE RANGE

$$\frac{1-x^2}{1+x^2} \quad (x \in \mathbb{R})$$

range

tangent-half-angle

rational

№42 The Unit Circle Parametrisation

●●●●● L3

PROVE THAT

$$\left(\frac{2t}{1+t^2}\right)^2 + \left(\frac{1-t^2}{1+t^2}\right)^2 = 1$$

parametrisation

unit-circle

identity

№43 A Rational Sine Equation

●●●●● L4

SOLVE

$$\frac{2x}{1+x^2} = a \quad (x \in \mathbb{R})$$

parameter

quadratic

range

№44 Linear Over Quadratic Peak

●●●●● L4

FIND THE RANGE

$$\frac{3x+4}{x^2+1} \quad (x \in \mathbb{R})$$

range

discriminant

substitution

№45 Twelve-One Peak

●●●●● L4

FIND THE RANGE

$$\frac{5+12x}{1+x^2} \quad (x \in \mathbb{R})$$

range

quadratic

extremum

№46 Unit Circle From a Line

●●●●● L3

SIMPLIFY

$$\sin(2 \tan^{-1} x)$$

inverse-tangent

double-angle

rational

№47 Cosine From Inverse Tangent

●●●●● L3

SIMPLIFY

$\cos \left(2 \tan^{-1} x \right)$

inverse-tangent double-angle rational

№48 Radical Touches the Line

●●●●● L4

SOLVE

$\sqrt{1-x^2} + x = 1 \quad (-1 \leq x \leq 1)$

radical sine-substitution extraneous

№49 Inverse-Sine Domain That Is All Real

●●●●● L4

FIND THE DOMAIN

$\sin^{-1} \left(\frac{2x}{1+x^2} \right)$

domain inverse-sine range

№50 Circle Parameter With Missing Point

●●●●● L4

EVALUATE

Find the point on $X^2 + Y^2 = 1$ represented by $X = \frac{1-t^2}{1+t^2}$, $Y = \frac{2t}{1+t^2}$ when $t = 1$.

parametric unit-circle substitution

№51 Radical Meets Quadratic

●●●●● L4

SOLVE FOR X

Solve for real x : $x + \sqrt{1-x^2} = \sqrt{2}(2x^2 - 1)$.

cosine substitution radical equation double angle extraneous roots

CHAPTER VI

Concavity, Tangents & Bounds



Concavity, tangent lines, Jensen and SOS bounds.

№52 The Basic Sharp Chain

●●●●● L3

PROVE THAT

$$0 < x < \frac{\pi}{2} \implies \sin x < x < \tan x$$

basic-inequality

area

tangent

№53 Triangle Sine Sum Ceiling

●●●●● L3

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

Jensen

triangle

sine

№54 Sine-Cosine Diagonal

●●●●● L3

FIND THE MAXIMUM

$$\sin x + \cos x \quad (0 \leq x \leq \frac{\pi}{2})$$

maximum

phase-shift

Cauchy

№55 Acute Tangent Jensen

●●●●● L4

PROVE THAT

$$A, B, C \in (0, \frac{\pi}{2}), A + B + C = \pi \implies \tan A + \tan B + \tan C \geq 3\sqrt{3}$$

Jensen

tangent

acute-triangle

№56 Cosine Sum Triangle Bound

●●●●● L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \cos A + \cos B + \cos C \leq \frac{3}{2}$$

triangle

cosine

sharp-bound

№57 Product of Sines Ceiling

●●●●● L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$$

product

Jensen

triangle

№58 Cosine-Square Floor

●●●●● L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \cos^2 A + \cos^2 B + \cos^2 C \geq \frac{3}{4}$$

cosine-square

triangle

minimum

№59 Sine-Square Ceiling

●●●●● L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$$

sine-square

triangle

maximum

№60 Two-Term AM-GM in Disguise

●●●●● L3

PROVE THAT

$$0 < x < \frac{\pi}{2} \implies \tan x + \cot x \geq 2$$

AM-GM

tangent

reciprocal

№61 Endpoint Tangent Line

●●●●● L5

PROVE THAT

$$0 \leq x \leq \frac{\pi}{2} \implies \sin x \geq \frac{2x}{\pi}$$

concavity

chord

sharp-bound

CHAPTER VII

Phase, Range & Extremum Machines

R

Phase shifts and envelopes find the range before calculus.

№62 Three-Four-Five Phase

●●●●● L3

FIND THE RANGE

$$3 \sin x + 4 \cos x$$

phase-shift

range

R-method

№63 Shifted Thirteen

●●●●● L3

FIND THE RANGE

$$2 + 5 \sin x - 12 \cos x$$

range

phase-shift

constant

№64 Quadratic Form on the Circle

●●●●● L4

FIND THE RANGE

$$\sin^2 x + 2 \sin x \cos x + 3 \cos^2 x$$

quadratic-form

eigenvalue

range

№65 Square-Difference Wave

●●●●● L4

FIND THE RANGE

$$4 \sin x \cos x + 3(\cos^2 x - \sin^2 x)$$

double-angle

range

phase

№66 Nonnegative Shift

●●●●● L3

FIND THE RANGE

$$5 + 3 \sin x + 4 \cos x$$

range

phase-shift

amplitude

№67 A Sinusoidal Equation Parameter

●●●●● L4

EVALUATE

Find all real k for which $\sin x + \cos x = k$ has a real solution.

parameter

range

phase-shift

№68 Sine Product Peak

●●●●● L3

FIND THE MAXIMUM

$\sin x \cos x$

- maximum
- double-angle
- product

№69 Secant-Tangent Square Range

●●●●● L4

FIND THE RANGE

$\frac{1 + \sin x}{1 - \sin x} \quad (x \neq \frac{\pi}{2} + 2k\pi)$

- range
- half-angle
- unbounded

№70 Quadratic Sine Range

●●●●● L4

FIND THE RANGE

$2 \sin^2 x - 2 \sin x + 3$

- quadratic
- range
- vertex

№71 Two-Variable Symmetric Peak

●●●●● L5

FIND THE MAXIMUM

$\sin x + \sin y \quad \text{given } x + y = \frac{2\pi}{3}$

- two-variable
- sum-to-product
- maximum

CHAPTER VIII

Periodicity & Root-Counting

Periodicity and intervals separate true roots from impostors.

№72 Double Sine Roots

●●●●● L3

SOLVE

$$\sin 2x = \sin x \quad (0 \leq x < 2\pi)$$

equation

root-counting

double-angle

№73 Cosine Double Roots

●●●●● L3

SOLVE

$$\cos 2x = \cos x \quad (0 \leq x < 2\pi)$$

equation

cosine

quadratic

№74 Triple Zero Count

●●●●● L3

SOLVE

$$\sin 3x = 0 \quad (0 \leq x < 2\pi)$$

periodicity

zero-counting

sine

№75 Quadratic Cosine Equation

●●●●● L3

SOLVE

$$2 \cos^2 x - 3 \cos x + 1 = 0 \quad (0 \leq x < 2\pi)$$

quadratic

cosine

interval

№76 Sine Five Meets Sine Three

●●●●● L4

COUNT

Number of solutions of $\sin 5x = \sin 3x$ on $[0, 2\pi)$.

counting

product-to-sum

frequency

№77 Cosine Plus Cosine Two

●●●●● L4

SOLVE

$$\cos x + \cos 2x = 0 \quad (0 \leq x < 2\pi)$$

equation

cosine

quadratic

№78 Sine Equals Cosine Double

●●●●● L4

SOLVE

$$\sin x = \cos 2x \quad (0 \leq x < 2\pi)$$

equation

sine

double-angle

№79 Tangent Reciprocal Equation

●●●●● L4

SOLVE

$$\tan x + \cot x = 4 \quad (0 < x < \frac{\pi}{2})$$

tangent

reciprocal

quadratic

№80 Zeros on a Closed Half-Period

●●●●● L3

COUNT

For positive integer n , count the roots of $\sin nx = 0$ on $[0, \pi]$.

counting

endpoint

periodicity

№81 Two-Equation Unit Sum

●●●●● L5

SOLVE

$$\sin x + \sin y = 1, \quad \cos x + \cos y = 1 \quad (0 \leq x, y < 2\pi)$$

system

sum-to-product

two-variable

№82 Where the Line Meets the Wave

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

Find the total number of real solutions of $\sin x = \frac{x}{10\pi}$.

bounding

graphical-count

line-vs-sine

symmetry

№83 Three Tangents in a Row

●●●●● L4

SOLVE FOR X

Solve $\tan x + \tan 2x + \tan 3x = 0$ for $x \in (0, \pi)$, stating how many solutions there are.

tangent

factoring

chebyshev

domain-restriction

№84 The Reachable Heights of k

●●●●● L4

FIND THE RANGE

Find all real k for which $\sin x + \cos x + \sin x \cos x = k$ has a real solution; and say how many solutions $x \in [0, 2\pi)$ occur when k is least.

parameter

substitution

range

solvability

№85 When the Sines Equal the Cosines

●●●●● L4

SOLVE FOR X

Solve $\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$ for $x \in [0, 2\pi)$, and count the solutions.

sum-to-product

factoring

mixed-equation

solution-counting

№86 A Product Pinned to One

●●●●● L5

FIND THE NUMBER OF SOLUTIONS

Find the number of solutions in $[0, 2\pi]$ of $\cos x \cos 2x \cos 3x = 1$.

product

extremal

bounding

solution-counting

CHAPTER IX

Triangle Geometry Through Trig



Sine rule, cosine rule, area, R , r and s through angles.

№87 Thirteen-Fourteen-Fifteen Radii

●●●●● L4

EVALUATE

For a triangle with sides 13, 14, 15, find R and r .

Heron

inradius

circumradius

№88 Included Sixty Degrees

●●●●● L3

EVALUATE

In $\triangle ABC$, $A = 60^\circ$, $b = 7$, $c = 8$. Find a and Δ .

law-of-cosines

area

included-angle

№89 Side From Circumradius

●●●●● L3

EVALUATE

If $R = 5$ and $A = 30^\circ$ in a triangle, find side a .

law-of-sines

circumradius

side

№90 Cosine Sum and Radii

●●●●● L4

PROVE THAT

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

triangle

radii

cosine

№91 Half-Sine Square Sum

●●●●● L4

EVALUATE

$$\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} \text{ in terms of } r, R.$$

half-angle

radii

triangle

№92 Right Triangle Inradius

●●●●● L3

PROVE THAT

$$\text{In a right triangle with legs } a, b \text{ and hypotenuse } c, \quad r = \frac{a + b - c}{2}.$$

right-triangle

inradius

area

№93 Half-Angle Tangent Length

●●●●● L4

PROVE THAT

$$\tan \frac{A}{2} = \frac{r}{s-a}$$

half-angle

inradius

tangent-lengths

№94 Radii Determine Cosine Sum

●●●●● L3

EVALUATE

If $R = 10$ and $r = 4$, evaluate $\cos A + \cos B + \cos C$.

radii

cosine-sum

triangle

№95 A Double-Angle Triangle

●●●●● L4

SOLVE

In $\triangle ABC$, $A = 2B$ and $a : b = \sqrt{3} : 1$. Find (A, B, C) .

law-of-sines

double-angle

triangle

№96 Three Sine-Rule Copies

●●●●● L3

EVALUATE

$$\frac{a}{\sin A} + \frac{b}{\sin B} + \frac{c}{\sin C}$$

sine-rule

circumradius

sum

CHAPTER X

Grand Hybrids

Cyclotomic sums, Chebyshev, substitution and inequalities fused.



№97 Chebyshev Root Filter

●●●●● L5

EVALUATE

$$\sum_{k=0}^{n-1} T_m \left(\cos \frac{2\pi k}{n} \right)$$

Chebyshev roots-of-unity filter

№98 Dyadic Product at a Sharp Angle

●●●●● L4

EVALUATE

$$\prod_{k=0}^{m-1} \cos \frac{2^k \pi}{2^{m+1}}$$

dyadic product exact-value

№99 Rational Range by Half-Angle

●●●●● L4

FIND THE RANGE

$$\frac{3 + 4t}{1 + t^2} \quad (t \in \mathbb{R})$$

substitution range discriminant

№100 Cyclotomic Chord With Parameter

●●●●● L5

EVALUATE

$$\prod_{k=1}^{n-1} \left(1 - 2a \cos \frac{2\pi k}{n} + a^2 \right) \quad (a \neq 1)$$

roots-of-unity product parameter

№101 Triangle Product Capstone

●●●●● L4

FIND THE MAXIMUM

$$A, B, C > 0, \quad A + B + C = \pi : \quad \sin A \sin B \sin C$$

triangle inequality product

№102 Derivative of the Root Filter

●●●●● L5

EVALUATE

$$\sum_{k=1}^{n-1} \csc^2 \frac{\pi k}{n}$$

roots-of-unity derivative cosecant

№103 Cosine Five Meets Cosine Three

●●●●● L4

SOLVE

$$\cos 5x = \cos 3x \quad (0 \leq x < 2\pi)$$

equation

product-to-sum

multiple-angle

№104 Triangle Side Projection

●●●●● L4

PROVE THAT

$$\text{In } \triangle ABC, \quad a \cos B + b \cos A = c.$$

geometry

sine-rule

conditional

№105 Chebyshev Bound

●●●●● L5

PROVE THAT

$$|x| \leq 1 \implies |T_n(x)| \leq 1.$$

Chebyshev

inequality

range

№106 Dyadic Product Equation

●●●●● L5

SOLVE

$$\prod_{k=0}^{m-1} 2 \cos(2^k x) = 1 \quad \left(0 < x \leq \frac{\pi}{2^m}\right)$$

dyadic

equation

telescope